



Exercise: Mechanical Repair Project (MySQL, PHP, HTML, CSS)

Imagine a small mechanical repair workshop that wants to keep track of its daily operations using a database. The workshop records information about customers (name, contact, address), the vehicles they bring for repair (plate number, model, year, type), and the repair jobs performed (problem description, repair date, cost). Each repair job is carried out by a mechanic, whose details (name, specialization, years of experience) must also be stored. A customer may own multiple vehicles, and each vehicle can have multiple repair jobs over time. Similarly, a repair job must always be linked to both a vehicle and the mechanic who worked on it. The goal of this project is for you to identify the different entities, their attributes, primary keys, and foreign keys, then create the database schema in MySQL. After creating the tables, you should practice writing a few CRUD (Create, Read, Update, Delete) queries. Finally, you should connect the database to a PHP script to demonstrate how applications interact with relational databases.

Tasks 1: Provide a Usecase Diagram of the project

Task2: Identify Entities and Relationships

- From the description above, list the entities (tables) you think should exist.
- For each entity, list its attributes (columns).
- Decide the primary key for each table.
- Identify the foreign keys that connect the tables together.

Task 3: Propose an appropriate Class diagram for the project

Task 4: Create Tables in MySQL

- Write SQL statements (CREATE TABLE ...) to create the database and its tables.
- Make sure to include primary keys and foreign keys.

Task 5: Insert Sample Data

- Write SQL queries to insert at least 3 customers, 5 vehicles, 3 mechanics, and 5 repair jobs.

Task 6: CRUD Queries

- Write an SQL query to insert a new repair job.
- Write an SQL query to retrieve all vehicles owned by a particular customer.
- Write an SQL query to list all repair jobs for a given vehicle.
- Write an SQL query to update the cost of a repair job.
- Write an SQL query to delete a vehicle record.

Task 7 : PHP Connection and Operations

- Write a PHP script to connect to your MySQL database.
- Using PHP and SQL together, perform at least two operations, for example:
 - - Insert a new customer into the database.
 - - Fetch and display all vehicles belonging to a particular customer.

Task8: HTML and CSS

- You have been provided with screenshots, and pictures of a sample website. Your task is to replicate the exact website layout using all the assets we have provided you with.
- You must build all the sections of the SINGLE page in the order **homePage.png -> aboutPage.png -> howItWorksPage.png -> whatOurClientSayPage.png -> whyChooseUsPage.png -> contactPage.png ->**

Note: Ensure to include your products in 3 D format (that is the three product imputed).